PRAJNA

Personalized Revision & Academic Journey Navigation Assistant

AI-Powered Exam Intelligence for NEET · JEE Main · JEE Advanced

40 Years of PYQs

8,000+ Questions

SLM + RAG Intelligence

v2.0 · 2026 · Open-weight models · Zero ongoing API cost

The Challenge Every Aspirant Faces

18M+

Students appear annually
for ~90,000 NEET seats

40 yrs

Of exam papers rarely
analysed systematically

59+

Chapters per student
no clear priority framework

What's Missing Today

- Students study from intuition — missing high-probability topics that repeat cyclically in PYQs
- No visibility into which chapters need 6h of focus vs. 30 min quick revision
- 40 years of PYQs contain clear patterns — but they're locked in scanned PDFs
- Teachers cannot deliver personalised guidance across hundreds of students simultaneously

System Architecture

PDF Papers
40 yrs · 3 exams

›

Claude AI Extract
One-time structured JSON

›

SQLite Database
8,000+ questions indexed

›

Prajna Intelligence
SLM + RAG + Predictions

›

Dashboards & API
Students · Teachers

One-time cost

Claude extracts questions once. All analysis runs fully offline — no recurring API costs.

Offline-first

SQLite + Python run on any local machine. No cloud dependency after extraction.

SLM-powered

Open-weight models (Ollama / HuggingFace) generate RAG-grounded insights locally.

Multi-exam

JEE Main, JEE Advanced, and NEET share one unified pipeline, DB schema, and API.

40 Years of Question Intelligence

NEET / AIPMT

1988–2025 · 37 years

~3,200 Qs

Biology · Chemistry · Physics

JEE Main

2002–2025 · 23 years

~2,800 Qs

Mathematics · Chemistry · Physics

JEE Advanced

1985–2025 · 40 years

~2,100 Qs

Mathematics · Chemistry · Physics

Structured Question Schema (per-question extraction)

```
{ "exam": "NEET",  "year": 2024,
  "subject": "Chemistry",
  "topic": "Electrochemistry",
  "micro_topic": "Nernst Equation",
  "difficulty": 3,  "marks": 4 }
```

8,000+	Total questions indexed
4 signals	Per question for ML difficulty model
100% offline	After one-time Claude extraction
3 tables	questions · topics · patterns (SQLite)

The Prajna Prediction Engine

```
score = (w_trend × freq) + (w_cycle × cycle_match)
      + (w_gap × years_since_last) + (w_cross × other_exam)
```

📈 Trend Analyzer

Topic frequency over time. Autocorrelation detects repeating cycles. Hot/cold identification.

🔄 Cycle Detector

Topics recurring every 3–5 years flagged with high cycle weight via time-series analysis.

⏱ Gap Scorer

Topics absent many years gain weight. 'Due to appear' heuristic from historical absence data.

🔗 Cross-Exam Signal

Topics appearing in both JEE and NEET weighted higher. Inter-exam correlation boosts confidence.

Student Priority Score:

```
priority = (100 − accuracy) × (slm_importance ÷ 100)
```

Critical	High	Medium	Low

The Prajna SLM Intelligence Layer

API Layer

FastAPI · 4 Routers

- GET /predictions/rank-all — global topic ranking
- POST /copilot/ask — conversational NL interface
- POST /reports/revision-plan — full study plan

Intelligence Layer

SLM + RAG Engine

- Ollama (Mistral 7B, Phi-3 — runs locally)
- HuggingFace (MPS/CUDA device support)
- ChromaDB vector store for PYQ evidence retrieval

Data Layer

Storage & Embeddings

- SQLite — 8,000+ structured questions
- PyTorch .pt embedding files per exam
- Metadata-filtered RAG retrieval pipeline

Anti-Hallucination Guards

All SLM outputs grounded in retrieved PYQ evidence. Confidence scores gated by source quality. Automatic fallback to pure prediction when RAG quality is low.

4-Stage Evaluation Pipeline

Grounding accuracy ≥ 85% · Factual consistency ≥ 75%
Ranking quality (Kendall τ) ≥ 0.70 · P95 latency ≤ 3,000ms
Fallback rate ≤ 10% · Insight usefulness ≥ 80%

Zero-Code Provider Switching

Single abstraction layer switches between Ollama, HuggingFace (MPS/CUDA), and OpenAI. Zero code changes — swap via .env. Mock provider for CI.

Student Intelligence Dashboard

A per-student window into performance, prediction alignment, and AI-guided revision — client-side, zero backend cost.

🏆 Hero Performance Card

Latest score, rank, improvement %, consistency score — all at a glance with trajectory context.

📈 Score Trajectory

Line chart across 10 mock exams. Per-exam trend line. Multi-subject accuracy overlay on one canvas.

📊 Chapter Performance

59 chapters grouped by subject. Sorted weakest-first with level badges M / S / D / W / C for each chapter.

🎯 Priority Focus Areas

Top 5 Prajna-ranked chapters: high gap AND high exam-appearance probability. Sorted by priority score.

⚡ Live Intelligence Feed

Real-time top-18 ranked topic predictions from the Intelligence API. Student weak-zone matches highlighted with ★.

🤖 Student Copilot

NL Q&A grounded in the student's actual performance data. Multi-turn memory with pre-seeded question suggestions.

Prajna-Powered Subject-Wise Study Guide

Interactive advisor for ALL chapters — not just top 5. Tells students what to study, how long, in what order, and why.

Subject Panel

Sorted by urgency (avg priority score desc)

Chapter Row

Ranked #1, #2... · dual bars: accuracy + importance

Chapter Detail Card

Study hours · Trend arrow · Gap to next mastery level

Micro-Topic Chips

Lazy API fetch per chapter · cached in Map · top 3 topics

Revision Strategy

5 levels × 3 trend directions = 13 unique templates

Priority Score Formula

```
priority = (100 − acc) × (slmImp ÷ 100)
```

Study Hours by Level

C → 6h W → 4h D → 2.5h S → 1h M → 0.5h

Urgency Thresholds

Urgent > 35 · Moderate 20–35 · On Track < 20

Lazy Loading + Cache

Micro-topics fetched only on chapter expand. Map prevents duplicate API calls.

Next Level Gap

gap = pctLevel() boundary − accuracy (C→25, W→45, D→65, S→80)

Development Roadmap

01 MVP API

3–5 days · ✅ Complete

- Prediction + ranking API (no SLM)
- FastAPI with 4 core routers
- Priority scoring + urgency tiers
- Topic cluster detection engine

02 SLM Insights

1–2 wks · ✅ Complete

- Open-weight SLM (Ollama / HF)
- RAG with ChromaDB vector store
- Anti-hallucination prompt guards
- Revision plan + trend endpoints

03 Copilot & Dashboard

1–2 wks · 🔵 Active

- Multi-turn conversational copilot
- Student-personalised recommendations
- Subject-wise interactive guide
- Teacher class-level intelligence

04 Eval & CI

2–3 wks · Planned

- Grounding accuracy ≥ 85%
- A/B prompt testing harness
- Feedback (thumbs up/down)
- Auto-regression in CI pipeline

Built on a Zero-Cost Stack

Component	Tool / Framework	Cost
Data Extraction	Claude AI (one-time)	Free tier
Database	SQLite	Free
ML Classifier	Scikit-learn + PyTorch	Free
Embeddings	Sentence Transformers	Free
SLM Inference	Ollama (Mistral 7B, Phi-3)	Free (local)
Vector Store	ChromaDB	Free
API Server	FastAPI + Uvicorn	Free
Dashboards	Streamlit + HTML / JS	Free

PRAJNA's mission: democratize exam intelligence. Every student — regardless of coaching budget — deserves the same data-driven edge previously available only to the most well-resourced.